

AI in Fraud Detection: Dankse Bank

Fraud detection is the process of identifying suspicious activity that indicates criminal theft of money, data, or resources. Fraud detection is important because of the costs and consequences businesses face without it. For example, financial loss, business interruption, loss of productivity, and reputational damage. As well as being required by law.

A lot of organizations have dedicated fraud prevention and detection teams that perform risk management assessments, assign risk scores to each fraud risk to determine which poses the greatest threats and should be prioritized, and then evaluates the fraud prevention measures and fraud detection solutions that it can use to address fraud threats based on their type and severity. But the industry is shifting towards AI, and many organizations like Dankse Bank now use AI to accelerate and improve their fraud detections systems.

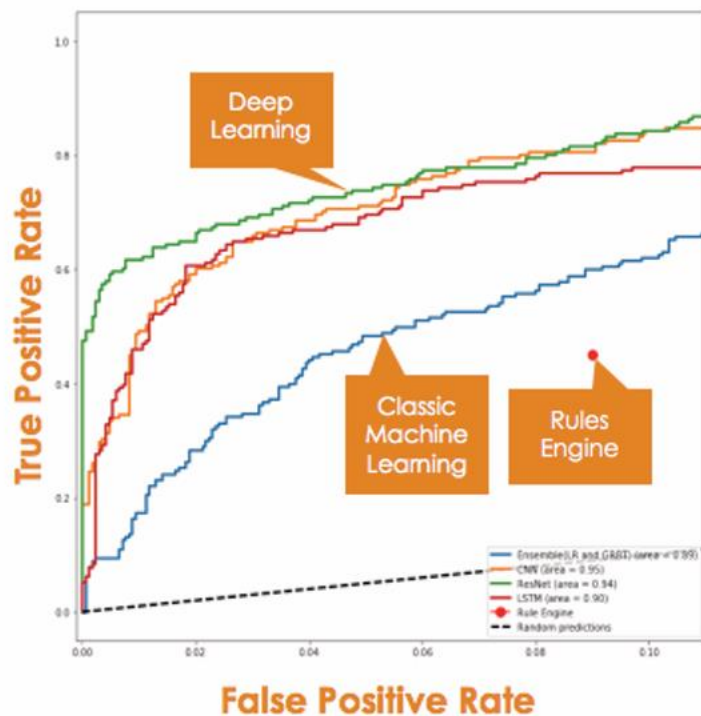
Dankse Bank is a Danish multinational banking and financial services corporation. It is the largest bank in Denmark and was founded in October 1871. Dankse bank needed to modernize its fraud detection defenses. They decided to collaborate with the Teradata Consulting company Think Big Analytics, that developed an AI-powered fraud detection platform. This platform analyzes thousands of hidden characters and evaluates millions of online banking transactions in real-time.

This system uses a champion/challenger approach, where the system runs multiple AI models (challengers) to learn from data. If one of the challengers is more accurate than the others, it becomes the champion, being the principal system while the others learn from it.

Before implementing this system, Dankse Bank was struggling with a low 40 percent of fraud detection and was managing more than 1,200 false positives per day. This large number of false positives required investment of money, time, people, compute power, etc. After implementing the AI fraud detection system in collaboration with Think Big Analytics they achieved:

- 60 percent reduction in false positives, with an expectation of reaching 80 percent.
- 50 percent increase in true positives.
- Notable reduction of fraudulent activities.

This graph shows how true and false positives rates improved (the red dot is the old rules).



Danske bank faced challenges when moving into deep learning and AI environment. This project required to gather and integrate quality data from emerging and existing sources, establishing pipelines for data processing, collaboration of bank representatives, engineers, data scientists.

Danske Bank is a great example of how to use AI to stop fraud. They achieved amazing results and showed other companies how to do it. This case study shows the benefits of AI, but also the challenges. For example, it is hard work to gather high-quality data from different sources while following legal and ethical rules. Also, it requires many different professionals to work together to make sure the system runs correctly. Thanks to AI, we are getting much better at detecting fraud. However, human supervision is still necessary for the most important decisions. Finally, the "champion/challenger" approach was a very interesting and important part of this case study, showing a new and smart way to keep improving technology.

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